

Power BI Data Modeling Project

This is a project that walks through turning a messy 23-table dataset into a clean Power BI star schema model that can be used reliably for reporting and analytics.

Project Goal

The objective is to transform a complicated dataset containing 23 interconnected tables into a well-structured star schema. This involves identifying business entities, separating facts from dimensions, creating relationships, building measures, and validating the final model.

Phases

1. Prepare & Explore (Step 1)
2. Create Dimensions (Steps 2 – 4)
3. Create the Facts (Steps 5 - 8)
4. Polish (Steps 9 – 12)

Rules

5. Build a Star Schema - A fact in the middle, dimensions around it.
6. Understand before making any changes – Always read the grain first.
7. Every column earns its place – If a column doesn't help the report, drop it.
8. Protect the numbers – Be aware of the Totals, re-check after every change.

Standards

1. Language: English
2. Naming: snake_case
3. Tables (prefix): fact_ & dim_
4. Keys (suffix): _id (From source) & _key (Created by us in PBI)
5. Friendly column names: customer_name, total_revenue, order_status

End-to-End Documentation of the Process

Step 1: Explore and Understand the Tables

Main Tasks

1. Inventory available tables.
2. Identify primary and foreign keys.
3. Understand business processes represented in data.
4. Detect redundant or unnecessary tables.

Step 2: Build the Customer Dimension

Main Tasks

1. Gather all customer-related attributes.
2. Remove duplicate customer records.
3. Create a unique customer key.
4. Structure attributes for reporting.

Example Attributes

- Customer ID
- Customer Name
- Region
- Country
- Segment

Step 3: Build the Product Dimension

Main Tasks

1. Combine product-related information.
2. Remove duplicates.
3. Create a clean product record structure.
4. Prepare products for slicing sales and inventory data.

Example Attributes

- Product ID
- Product Name
- Category
- Brand
- Subcategory

Step 4: Distinguish Facts vs Dimensions

Fact Tables

Contain measurable business events:

- Sales
- Orders
- Inventory movements
- Campaign performance

Dimension Tables

Contain descriptive information:

- Customer
- Product
- Date
- Geography

Step 5: Build the Sales Fact Table

Main Tasks

1. Bring transaction records together.
2. Keep sales metrics.
3. Link transactions to dimensions.

Example Measures

- Revenue
- Quantity Sold

- Discount
- Profit

Result

A transactional table that becomes the foundation of sales reporting.

Step 6: Build the Inventory Fact Table

Main Tasks

1. Model inventory positions.
2. Track stock movements.
3. Link inventory records to products and dates.

Example Metrics

- Stock Level
- Inventory Value
- Reorder Quantities

Step 7: Build the Campaign Fact Table

Main Tasks

1. Store campaign performance metrics.
2. Connect campaigns to customers and products.
3. Support marketing analysis.

Example Metrics

- Campaign Cost
- Conversions
- Clicks
- Revenue Generated

Step 8: Build the Order Fulfillment Fact

Main Tasks

1. Capture order processing activities.
2. Track fulfillment status.
3. Measure delivery performance.

Example Metrics

- Orders Fulfilled
- Delivery Time
- Shipment Count

Step 9: Finish the Remaining Model

Main Tasks

1. Connect all tables.
2. Remove unnecessary relationships.
3. Verify cardinality.
4. Optimize structure.

Step 10: Create Date Table and Core Measures

Date (Calendar Table) Attributes

- Year
- Quarter
- Month
- Week
- Day

Example Measures (DAX)

- Total Sales

- Total Orders
- Profit
- Average Order Value

Step 11: Implement Row-Level Security (RLS)

Main Tasks

1. Create security roles.
2. Restrict data visibility.
3. Test user access.

Benefits

- Protects sensitive data.
- Allows region-specific access.
- Supports enterprise deployments.

Step 12: Final Validations

Validation Checklist

- Relationships work correctly.
- Filters propagate properly.
- Measures return expected values.
- Security functions correctly.
- Performance is acceptable.

Final Outcome

The original 23-table "nightmare" model is transformed into a structured star schema with:

- Clean dimensions
- Well-defined fact tables
- Consistent relationships
- Reusable measures
- Security implementation
- Better reporting performance

